

Project Prometheus

Team 71 Project Technical Report to the 2024 Spaceport America Cup

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Princeton Rocketry constructed a 12 foot, 11 inch long 5.5 inch diameter fiberglass rocket weighing 77.75 pounds fully loaded. The rocket will reach an apogee of about 30,000 feet, powered by an Aerotech O5500X motor. At apogee, a 48" drogue chute will slow the rocket to ~65 ft/s. At 1,000 feet AGL, a 120" main chute will deploy to slow the rocket further to 16.4 ft/s. Both deployment events will be triggered by our Telemega flight computer, as well as our redundant Easymini altimeter, which will trigger both CO₂ (drogue) and black powder (main) ejection charge canisters to break shear pins holding airframe sections together. Our Telemega flight computer will also serve as our GPS tracker. Our payload will feature a CanSat and a 1U CubeSat fitted with various sensors to monitor rocket progress and internal environment, and a 2U Cubesat to study battery degradation mechanisms in space missions. To validate the performance of the flight vehicle, we simulated the rocket in OpenRocket and AeroFinSim and conducted the following tests: ejection charge sizing, recovery system deployment tests, avionics pressure tests, e-match ignition tests, and battery drainage tests.

I. Nomenclature

g	=	gravitational acceleration ($\sim 9.81 \text{ m/s}^2$)
"	=	inches
'	=	feet
s	=	seconds
LCO	=	Launch Control Officer
MCO	=	Mission Control Officer
MCC	=	Mission Control Center
RSO	=	Range Safety Officer
CATO	=	Catastrophic failure, usually of the propulsion system
APCP	=	Ammonium perchlorate composite propellant
COTS	=	Commercial off the shelf
IREC	=	Intercollegiate Rocket Engineering Competition
GPS	=	Global positioning system
MSL	=	Mean sea level

II. Introduction

Since Princeton Rocketry was founded in 2015, the club has made large strides to make rocketry and other aerospace-related hands-on projects available to Princeton students. The club supports L1 and L2 certifications, is designing and building a rocket to reach the Kármán Line, flies high-altitude balloons, and participates in two NASA competitions, among other projects. Princeton Rocketry has previously competed in the Spaceport America Cup on two occasions: 2019 (10k COTS) and 2023 (30k COTS). Last year's project (Project Star Scream) was the club's first attempt at launching a rocket to 30,000 feet. While we had a successful launch, we did unfortunately encounter a GPS failure, meaning that we were unable to recover the rocket and received zero points in the flight

category. Back and hungry for more, we are once again flying in the 30k COTS division, building off of both our successes and failures from last year.

This year's rocket was designed completely from scratch beginning in September 2023, with all of the lessons we learned at the 2023 Spaceport America Cup kept in mind as design decisions were being made. Far into our manufacturing process, we were forced to make an emergency design change when we received word that our original motor order for a Cesaroni O3400 would not be ready in time for the competition. The motor distributor we had placed the order suggested that we switch to an AeroTech O5500X, which they guaranteed was in stock. After re-running all simulations on the vehicle, we decided to make the switch, and had to lengthen the airframe by an extra 10 inches to accommodate the larger motor.

Princeton Rocketry has 76 registered members, with 33 of them being members of the Spaceport America Cup team. The team is composed of 19 Mechanical and Aerospace Engineering students, 7 Electrical and Computer Engineering students, 4 Computer Science students, 1 Physics student, 1 Operations Research and Financial Engineering student, and 1 Public and International Affairs student. Two team members are PhD candidate graduate students, while the rest are undergraduate students. The team is led by Team Lead Jake Vazquez '26, and is split into five smaller sub-teams: Airframe, Avionics/Telemetry, Payload, Recovery, and Operations & Logistics. The sub-teams are led by Simon Ji '28 and David Wang '27 (co-leads), John Wallace '25, Kendall Olszowka '26, Maia Abiani '26, and Tate Keuler '26, respectively. We are additionally supported by student officers of Princeton Rocketry as well as our Level 3 mentor Michael X. Infante. We typically meet once a week for two hours during the semester and hold ad hoc build sessions to work on the flight vehicle.

The process of designing, building, and testing Project Prometheus has provided, and will continue to provide us with invaluable data and lessons learned for Princeton Rocketry's SpaceShot project, which is slated to launch in May 2025. SpaceShot is a multi-year project with the goal of being the first ever student-built two-stage rocket to reach the Kármán Line - the internationally recognized boundary of space. Many of our current Spaceport America Cup team members will be transferring to the SpaceShot team at the beginning of the 2024-2025 academic year, supporting the engineering, manufacturing, testing, and launch stages of the project.

III. System Architecture Overview

A. Integrated System Overview



Fig. 1 Cutaway view of flight vehicle.

Our flight vehicle is designed to be safe, reliable, and redundant. Our airframe has a 5.5-inch diameter and is 155 inches long. It will weigh approximately 78 lbs after loaded with propellant. The vehicle is divided into three main sections: nosecone, upper airframe, middle airframe, and lower airframe, all made of G12 fiberglass from Madcow Rocketry. All bulkplates were manufactured in our on-campus machine shop from two sheets of 1/8" G10 fiberglass. The nosecone will house the 48" drogue parachute and the associated CO₂ gas system used to deploy the parachute. Following the nosecone is the upper airframe, which will house the vehicle's payload bay. The 5-CubeSat-Unit payload is secured within the payload bay using two 1/4" threaded rods. Between the upper and middle airframes is a G12 fiberglass coupler that contains the avionics and electronics bay, housing the two altimeters, GPS system, and an on-board camera. The middle airframe will house the 10' main parachute and the associated black powder charge system used to deploy the parachute. Finally, the lower airframe will house the Aerotech O5500X motor system. All fiberglass bonds are secured with West Systems epoxy.

Four trapezoidal fins on the lower airframe will stabilize the rocket during flight. While our simulations suggest it was not entirely necessary, we completed a three-ply carbon fiber layup over the fincan to reinforce the fins during launch and landing, just for extra safety and redundancy.

In terms of recovery, 12' of 2100 lb $\frac{5}{8}$ " tubular nylon shock cord and 25' of 2800 lb braided kevlar shock cord connect the nosecone and upper airframe as well as the upper and lower airframes after chute deployment.

The vehicle is painted primarily white to reduce heating from solar radiation, with an orange nosecone for visibility purposes. The vehicle's flight profile is that of a standard dual-deployment rocket, beginning with motor ignition and boost, followed by an unpowered coast phase, the deployment of the drogue parachute at apogee, and ending with the deployment of the main parachute at 1,000' and vehicle touchdown.

B. Propulsion

The vehicle uses a Aerotech O5500X single use solid rocket motor, fuelled by the proprietary AeroTech's Propellant X. From Ref. [1], the motor has an average thrust of 5,514.6 N and a maximum thrust of 7,551.5 N. It has a specific impulse of 129.88 seconds and a burn time of 4.0 seconds. This rocket motor was chosen so that enough impulse can be given to the rocket in the time between ignition and launch rail departure to ensure stability, and to provide enough thrust to reach the desired maximum altitude of approximately 30,000 feet. The motor is plugged and has no ejection charge.

The motor casing is provided with the motor by AeroTech and distributor Animal Motor Works. The motor casing is secured using an Aeropack 98 mm Flange Retainer, which is adhered to the inner motor tube in the lower airframe using epoxy.

C. Aero-structures

1. Nosecone

Our nosecone is a 24" COTS filament-wound fiberglass nosecone with a precision-machined aluminum tip from Madcow Rocketry. The nosecone shoulder is 8.25" long and secured to the nosecone with six steel 4-40 screws. Inside the nosecone, we bonded using epoxy a fiberglass bulkplate (consisting of two $\frac{1}{8}$ " thick G10 bulkplates bonded together). A $\frac{3}{8}$ " eye bolt is secured to the bulkplate using a $\frac{3}{8}$ " nut and washers for load transfer. The drogue shock cord attaches to this eye bolt via a 5/16" 316 steel quick link. Two CO2 ejection canisters are attached to this bulkplate with steel 4-40 screws. The drogue parachute and shock cord are stored inside the nosecone shoulder. Three $\frac{1}{4}$ " vent holes are drilled 3" below the top of the nosecone shoulder to allow pressure to equalize inside the rocket during boost and coast.

2. Upper Airframe

The upper airframe is a COTS 36" long G12 filament-wound fiberglass cylinder with a 5 $\frac{3}{8}$ " inner diameter from Madcow Rocketry. It is secured to the nosecone shoulder with three 4-40 nylon shear pins during boost and coast. The shear pins break at apogee to release the drogue parachute. The upper airframe also contains the payload bay and the avionics/electronics bay (in a coupler between the upper and middle airframes). The payload interfaces with the rocket via 2 $\frac{1}{4}$ " steel threaded rods, which all loads pass through. Three $\frac{1}{4}$ " vent holes are drilled into the upper airframe to allow pressure to equalize inside the rocket during the flight.

A coupler at the bottom of the upper airframe houses our avionics/electronics bay, which contains a Telemega flight computer, a backup Easymini flight computer, and an onboard GoPro Hero 11 Mini camera. These avionics are powered by two 3.7-volt lithium-ion batteries. The avionics are contained within a 3D-printed Nylon housing, which slides onto two $\frac{1}{4}$ "-28 threaded rods. These rods are bolted onto two G10 bulkplates. The bottom bulkplate is bonded to the top of the coupler with West Systems epoxy, while the top bulkplate remains unbonded to allow access to the avionics. Each bulkplate has a $\frac{3}{8}$ " eye bolt where shock cords attach via a quick link. The bottom bulkplate has two black powder ejection canisters attached with $\frac{1}{4}$ "-20 screws for main parachute deployment.

3. Middle Airframe

Following the avionics/electronics bay is the middle airframe, which is a COTS 18" G12 filament-wound fiberglass tube from Madcow Rocketry 5 $\frac{3}{8}$ " inner diameter. Within this section resides the main parachute and its shock cord, which are connected to the avionics bay bulkplate and the motor casing with $\frac{3}{8}$ " eyebolts and quick

links. Three $\frac{1}{4}$ " vent holes are drilled into the middle airframe to allow pressure to equalize inside the rocket during the flight.

4. Lower Airframe

The lower airframe consists of a 70" long G12 filament-wound fiberglass tube from Madcow Rocketry with a $5\frac{3}{8}$ " inner diameter. The lower airframe contains the motor retainer, the motor casing, and the AeroTech O5500X motor. Four $\frac{3}{8}$ " fins are mounted to the airframe at 90 degree intervals. A 48" long, 4" diameter COTS G12 filament-wound fiberglass inner motor tube from Madcow Rocketry is attached inside the lower airframe using epoxy and three centering rings. Each fin runs into the airframe, and is epoxied to both the inner and outer tubes for extra strength, with a $\frac{1}{8}$ " fillet on each side. The fillet is made of West System 206 epoxy resin mixed with fiberglass microsphere thickener by Fasco Epoxies Inc. To reinforce the fin can, we completed a three-ply carbon fiber tip-to-tip layup using 2x2 twill carbon fiber and West System 206 epoxy resin. The first layer extends $\frac{1}{3}$ of the way up the fins and is oriented at 45 degrees. The second layer extends $\frac{2}{3}$ of the way up the fins and is oriented at 135 degrees. Finally, the third layer is tip-to-tip and is oriented at 90 degrees. The stiffness and thickness of the fins is meant to resist aerodynamic flutter, divergence, and aerodynamic moments throughout boost and coast phases of flight. Located 6.5" and 70.5" from the bottom of the lower airframe are two COTS 1515 rail buttons, attached to the frame using $\frac{1}{4}$ " screws. There is one rail button located on either side of the center of gravity of the vehicle. The bottom of the lower airframe features a 5.5" long boattail, which brings the airframe diameter from 5.5" to 4" for aerodynamic efficiency. Three $\frac{1}{4}$ " vent holes are drilled into the boattail to prevent propulsion inefficiencies related to the Krushnic Effect.

D. Recovery

1. Overview of Deployment Systems

Two different deployment systems are being employed for both the drogue and main parachutes. For the drogue parachute deployment the rocket will be using a CO2 energetic system at ~30,000 feet. This was chosen due to the unreliability of black powder burning at high altitudes; Black powder begins to burn inconsistently at altitudes exceeding 20,000ft. For the main parachute deployment at 1,000 feet the rocket will be using a black powder energetic system on the basis that at 1,000 feet AGL black powder does not have similar combustion issues, and it is more economically efficient than CO2 for this use.

2. Flight Computer System

The flight computer system consists of two distinct flight computers, a Telemega and Easymini. The Telemega includes a GPS tracker in addition to barometers, which measures altitude AGL after an initial ambient pressure calibration reading. This information is transmitted to a ground station over a ham-band radio link. The Easymini is barometric only, and measures altitude similarly. Both computers are powered by separate lithium ion 3.7 V batteries (provided by flight computer manufacturer) and screw switches. These screw switches will be accessed through the electronics bay vent holes. The altimeters are mounted in the main payload bay on a Nylon/glass fiber composite sled using 4-40 mounting hardware. The batteries are held in enclosures and zip-tied in.

Both altimeters contain leads to two Firewire e-matches, one for the drogue deployment event and one for the main deployment event. The CO2 energetic system still requires a very small amount of black powder, which is completely sealed off from outside of air in the housing, which is why an e-match is still required for the drogue deployment. For the main deployment, the e-matches are contained in separate black powder wells to maximize redundancy. The main well will contain 7.5 grams of black powder for the primary and 9 grams for backup. The quantity of black powder was determined using an online ejection charge calculator [2] and validated via a separation ground test.

3. *Primary GPS Tracker*

The primary GPS tracker is the Telemega flight computer, powered by a 3.7V 850 mAh battery. It is mounted on the electronics bay sled, and will be powered during vehicle assembly. It is intended to be used to track the rocket during flight and record its touchdown location to ensure a smooth recovery.

4. *Parachutes*

The drogue parachute is a Fruity Chutes 48" ripstop elliptical parachute with 400 lb braided nylon shroud lines, 5/8" nylon bridle and 1500 lb swivel. It is intended to reduce the vehicle's velocity to approximately 45 ft/s during the main descent. The drogue parachute is red and white. The drogue chute is protected from the ejection charge using a parachute liner from Fruity Chutes as well as a Nomex blanket.

The main parachute is an Iris Ultra 120" standard nylon parachute with 400 lb paroline shroud lines and a 3000 lb swivel. It reduces the vehicle's velocity to approximately 16 ft/s during its final descent and ground impact to prevent damage to the airframe. The main parachute has primary colors of orange and black. The main chute is protected from the ejection charge by a parachute deployment bag from Fruity Chutes as well as a Nomex blanket.

5. *Recovery Harness*

We sized our recovery system such that each component could take at least 2,500 lb load. This number was determined by multiplying the vehicle's weight after motor burnout by 45 g. Each shock cord is composed of 12" of 2100 lb $\frac{5}{8}$ " tubular nylon shock cord as well as 25" of 2800 lb braided kevlar shock cord. The drogue shock cord is attached to the 1300 lb $\frac{3}{8}$ " eye bolt on the nosecone via a 2400 lb quick link. Note that while the eye bolt has a 1300 lb working load, the breaking load is at least two times higher, fitting into our safety factor. The other side of the shock cord is attached to the eye bolt at the top of the upper airframe. The main shock cord is attached to an eye bolt via a quick link on the avionics/electronics bay on one side, and to the $\frac{3}{8}$ " eye bolt screwed into the forward closure of the motor casing on the other side via a quick link. We expect the long shock cord to mitigate the high torsional forces that would require a swivel.

E. Payload

1. *Mission Requirements*

The purpose of the payload is to get data about the rocket location and internal environment, as well as to run a scientific experiment that explores battery degradation mechanisms in space missions. It was determined that a non-deployable CubeSat structure would be best for this mission. With the given payload bay space, a 5U (10x10x50cm) Cubesat was initially decided upon, but this space was soon downgraded to 4U to account for the extra space the recovery systems would need. The team decided upon breaking the payload into three components that would be connected via rods and stay as a unit inside of the rocket body. This consisted of a CanSat and two CubeSats (1U and 2U), each of which would house their own sensors and scientific experiments.

2. *CanSat Overview*

The CanSat is made out of Aluminum 7075 and contains four round layers that were initially decided to be around half an inch thick. Machining proved itself to be a bit difficult when it came to cutting, so these layers ended up not all being of the same thickness. These layers were connected with steel rods and bolted to leave adequate space between each layer. On each layer of the CanSat, sensors from MaxIQ payload kits were mounted. These sensors are able to monitor air pressure, oxygen levels, rocket acceleration, and weather. The chips were mounted above small plastic spacers to ensure that the metal to metal contact of the chips to the CanSat would not affect their functionality. Two batteries are also mounted on two of these levels to power the MaxIQ chips. The data recorded with these sensors is saved on an internal SD card and must be retrieved after the mission.

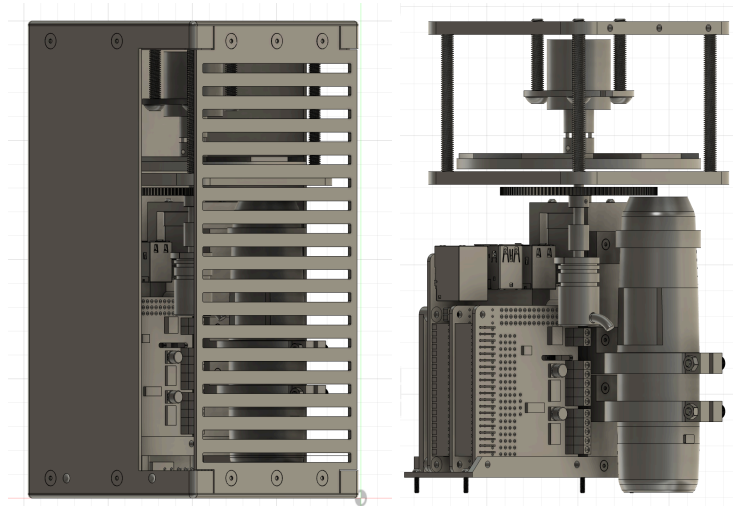
3. *1U CubeSat Overview*

The 1U CubeSat's primary use is to hold the Spot GPS Tracker that serves as a backup GPS on our mission. The structure was designed in CAD and 3D printed out of PLA into three pieces. There is the main structure, which

contains holes for rods to hold the CubeSat assemblies together and a box-like structure with hollow walls. There are also two identical top and bottom pieces which are solid and have a set of holes for the rods to hold the CubeSats together, a set of holes for the rods that hold the whole payload together, and a set of holes for the rods to hold the payload to the rocket. The Spot GPS Tracker is mounted onto the bottom piece.

4. *2U CubeSat Overview and Experimentation*

The 2U CubeSat is 3D printed from PLA and includes an experiment designed by Simon Ji, a first-year PhD student at Princeton, in collaboration with the Hatzell Lab. The goal of the experiment is to study battery degradation mechanisms in space. The first part of the experiment will test the batteries' response to high-G forces by storing several commercial Li-ion rechargeable coin cell batteries in the payload bay, which will then be cycled through a galvanostat battery cycler at Hatzell Lab in Princeton and its data compared with a control data set. The second part of the experiment will be to test the electronic systems required for an upcoming CubeSat mission which hopes to explore dendrite growth in batteries under microgravity. While the Spaceport America Cup flight cannot achieve microgravity due to the lack of active controls, it is still very valuable to verify the validity of design, which will be useful for future rigorous reviews by NASA. This experiment was built in collaboration with NanoRacks and Princeton TigerSATS, and required a custom PCB board printed and soldered by PCBWay to conduct in-situ electrochemical experiments. The system will be controlled by a Raspberry Pi and powered by 10 AA batteries placed in series to achieve the 15 volts required for the electrochemistry. During the descent phase (right after apogee to before main chute release), the automated system will turn on and inject current into a thin battery tape cell, and a video will be recorded through an on-board microscope.



Mock-ups of the Electrodeposition Experiment

5. *Construction*

Meeting weight and strength requirements pushed the use of an aluminum CanSat and a steel rods to hold the CubeSats together, the payload assembly together, and the assembly to the rocket. In using extra sets of rods, we were able to ensure that our payload was strong and secure. By printing the CubeSats out of PLA, we could find ways to make our payload lighter. Machining the CanSat out of metal also helped to ensure that our MaxiQ sensor chips would be protected on a solid surface.

IV. Mission Concept of Operations Overview

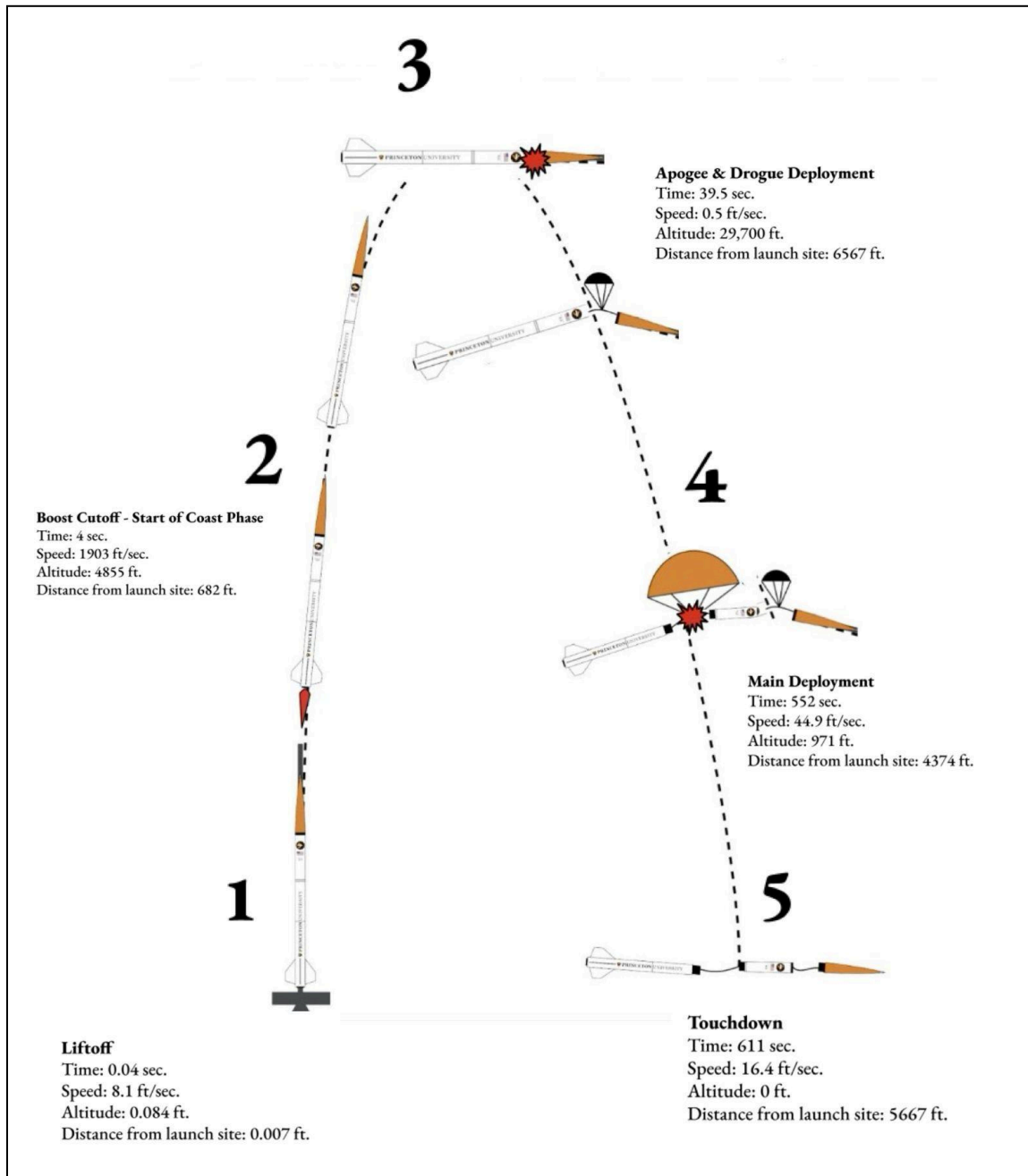


Fig. 2 Concept of Operations Graphic

A. Assembly

In the assembly phase, the vehicle will be horizontally integrated, starting with non-critical payload and ending with motor and electronics bay assembly. All electronics will be powered down at the start of this phase. First, the 4U CubeSat payload (consisting of two CubeSats and a CanSat) will be assembled and powered. The drogue parachute and shock cord will be packed and attached to the nosecone eye bolt and the eyebolt at the top of the upper airframe. The two sections will be secured with shear pins. Once secured, the CO₂ ejection system will be installed, shortly followed by the installation of the payload on the two 1/4" steel rods that run through the airframe. Next, the middle and lower airframe sections will be assembled, and then the motor will be assembled and integrated with the lower airframe. The main parachute will be packed and the recovery harness attached to the

forward closure of the motor. Following parachute attachment, the lower and middle airframes will be secured via shear pins. Finally, the avionics/electronics bay will be assembled. The flight-critical electronics will undergo final checks, and the bay will be assembled with the altimeters powered on but without igniters connected. Then, power will be connected to the altimeters but the mechanical switches will remain off. Igniters will also be connected. The coupler containing electronics bay will be integrated into the upper and lower airframes to complete flight vehicle assembly, secured using 4-40 steel screws.

B. Preflight

In the pre-flight phase, the vehicle will be carefully loaded onto the rail with the motor igniter removed and recovery electronics unarmed. Then, the mechanical screw switches for the flight computers will be powered on. The rail will then be raised, and after final vehicle checkout non-critical personnel will exit the vicinity. The recovery electronics will be armed, and the motor igniters inserted.

C. Launch

1. Countdown

In the countdown phase, the team will observe range rules regarding the countdown process by the Launch Operations Lead. This phase will end when the FIRE signal is sent to the igniter.

2. Ignition and Liftoff

The ignition and liftoff phase begins when the FIRE command is sent to the igniter, which will then begin the ignition sequence of the motor. When the motor comes up to chamber pressure and the vehicle begins to move up the rail, the two rail buttons will ensure the vehicle remains oriented vertically. The motor will increase the vehicle's velocity to ~121 ft/s, ensuring its stability upon leaving the rail (~1.5 calibers). Upon the exit of the first rail button from the rail, the ignition and liftoff phase ends.

3. Boost

When the first rail button leaves the rail, the vehicle will be stabilized by aerodynamic forces. The stability of the rocket will not fall under 1.5 body calibers to ensure this. During the 4.0 s burn time of the motor, the vehicle will undergo ~24 g's of acceleration, which the airframe structure has been designed to withstand. Upon sensing acceleration, the flight electronics will register the launch and await apogee. All parts of the rocket are intended to remain intact during boost. Upon motor burnout, this phase will conclude.

4. Coast

During the coast phase, the rocket will undergo deceleration due to gravity and drag forces. To prevent internal pressure or drag separation forces from demating any part of the rocket, 3 nylon shear pins are located at the nosecone joint, and 6 nylon shear pins are located at the lower airframe / middle airframe joint - both of which are intended to separate later in the flight. These pins are intended to remain intact during the coast phase. Joints intended to remain connected during the flight are connected with 4-40 steel screws. During coast, the Telemega flight computer will transmit the vehicle's location and velocity to the team. Upon reaching apogee, the coast phase will conclude.

5. Initial Descent

The initial descent phase begins upon the rocket reaching apogee. At this point, the joint between the upper airframe and the nosecone will split. This will be accomplished by the primary flight computer firing the 16g CO₂ canister, pressurizing the compartment and breaking the shear pins, separating the airframe and deploying the drogue parachute. Two seconds after reaching apogee, the secondary flight computer will fire a backup 23g CO₂ canister to ensure drogue deployment in the event of a failure of the main flight computer. The two second margin was chosen to give ample time between the two charges in case the primary altimeter registers apogee late, and to allow time for the nosecone to completely eject before the secondary charge fires in a nominal deployment sequence. The drogue

parachute will then slow the vehicle's descent to ~45 ft/s. In order to prevent main parachute deployment, the shear pins connecting the lower body to the upper body tube will remain intact. During this phase, the GPS tracker will continue to update the team on the rocket's location and velocity. This phase ends upon the rocket reaching an altitude of 1,000 ft AGL.

6. Final Descent

Upon reaching an altitude of 1,000 ft AGL, the final descent begins. The main flight computer will ignite a charge containing 7.5 grams of black powder to pressurize the lower airframe, breaking the shear pins, and ejecting the upper body and main parachute. The main parachute will slow the vehicle's descent to about 16 ft/s. Upon reaching an altitude of 800 ft AGL, the secondary flight computer will ignite a charge containing 9 grams of black powder to ensure main parachute deployment in the event of a primary flight computer failure. Throughout this phase, the GPS tracker will continue to transmit the rocket's location to the team. This phase ends upon all parts of the rocket reaching 0 ft altitude and 0 ft/s velocity (touchdown).

7. Recovery

Upon touchdown, the team will use the GPS tracker to locate the rocket. Following Spaceport America Cup regulations, the team will depower the flight computers, remove the fractured shear pins, and reassemble the rocket. Upon returning to the competition area, the payload bay will be disassembled, all electronics will be powered down, and the motor casing will be cleaned. The flight data will then be collected from both flight computers and analyzed. Upon all electronics being depowered, the motor being cleaned, and the rocket reintegrated for transport, this phase is completed, as is the mission.

V. Conclusion and Lessons Learned

A. Conclusion

Princeton Rocketry designed and built a flight vehicle designed to reach 30,000 ft on an O-class motor, composed mainly of COTS components - making Project Prometheus the largest and most powerful rocket Princeton Rocketry has ever built for the Spaceport America Cup. This is the third time Princeton has competed in the annual competition, but the first time a majority of our current team members have built a rocket of this size and scale. As a result, we were in constant communication with our mentor, conducted our own research, and consulted with notes left from last year's Project Star Scream to design and manufacture our rocket. The learning process, alongside time and classwork constraints, proved to be our greatest challenges during the construction of the project. While challenging, Prometheus has been an extremely rewarding project, and has provided each team member with an invaluable amount of knowledge and experience.

B. Lessons Learned

1. Aero-structures

Our team gained experience working with carbon-fiber layups. Only one of our team members had ever done a composite layup before (for last year's Project Star Scream vehicle), so we conducted research by consulting our mentor, watching videos, and consulting with members of last year's Spaceport America Cup team. The experience proved invaluable to many team members, and will be incredibly helpful in building future launch vehicles.

2. Recovery

The team gained experience with designing recovery systems, sizing parachutes and shock cords, as well as conducting ejection charge tests. For instance, we learned to use a combination of nylon and kevlar shock cords for strength and flexibility. Furthermore, the team also gained valuable experience and knowledge working with a CO₂ ejection system for the first time.

3. Scientific Payload

The requirements prompted unique lessons about mechanical design and manufacturing. Meeting weight and strength requirements pushed the adoption of aluminum housing for the CanSat, while the use of lightweight PLA for the CubeSats allows for a quick iterative design process on a complex payload; PLA is also much cheaper and it was not necessary to replace it with stronger material since the payload housing will not be taking any direct load.

For the electrodeposition experiment, we gained valuable experience in engineering design that has multiple strict requirements. We came across many challenges while attempting to balance the scientific goal and realistic engineering issues. We also learned useful skills especially in electronic design, custom PCB printing and Raspberry Pi programming.

4. Team Management, Organization, and Logistics

This year's Spaceport America Cup team has seen a lot of growth in comparison to last year's team, which had a total of six members. Having a much larger team of thirty three this year, while easier to delegate tasks, provided a variety of new logistical challenges. With more team members, all with differing schedules, scheduling large full-team build sessions became almost impossible, especially as the semester progressed and workloads increased. To complete all components of the rocket by the end of the semester, the individual sub-teams had to schedule their own meetings in order for all team members to be able to contribute.

The organizational structure of the team this year, with one team lead and six sub-team leads, typically worked very well in terms of keeping the project on-track and managing the flow of information. An agenda was prepared for each team meeting we held, and tasks were managed through a task-managing spreadsheet. This helped to keep team members motivated and kept our member retention rate high throughout the year. We will likely continue to follow this structure in the future.

Planning travel logistics was also much more complicated. This year, we are bringing a team of 13 to New Mexico - a stark contrast from the 3 we brought last year. With team members scattered across the country, planning affordable flights that all arrived around the same time became very challenging. Additionally, a larger team required more equipment to be purchased and transported from Princeton - thus, the decision was made to send a team of four members two days earlier than the rest of the team to drive the rocket and other equipment across the country to New Mexico. Finding the stops for the drive, as well as overnight lodging, required an ample amount of planning to ensure we would reach our destination on time.

System Weights, Measures, and Performance Data Appendix

A. Basic Rocket Information

Table 1 Basic Rocket Information

Number of stages	1
Vehicle length	155"
Airframe diameter	5.5"
Number of fins	4
Fin semi-span	5"
Fin tip chord	4"
Fin root chord	18"
Fin thickness	0.375"
Vehicle weight	40.75 lb
Propellant weight	21.56 lb
Empty motor case weight	15.78 lb
Payload weight	8.8 lb
Liftoff weight	77.75 lb
Center of pressure	109" from forward end
Center of gravity	93.56" from forward end

B. Propulsion Information

Table 2 Propulsion Information

Motor type	COTS Single-Use
COTS manufacturer and designation	Aerotech O5500X
Motor letter classification	O
Average thrust	5,514.6 N
Total impulse	21,384.0 N-s
Motor burn time	4.0 s

C. Predicted Flight Data

Table 3 Predicted Flight Data

Launch rail length	140.5" (from top of ESRA-provided rail to rail guide)
Liftoff thrust-weight ratio	17.48

Rail departure velocity	121 ft/s
Minimum static margin	1.5 calibers
Maximum acceleration	762 ft/s ²
Maximum velocity	1,958 ft/s
Fin flutter velocity	Mach 5.29
Target apogee	30,000'
Predicted apogee	29,700'

D. Flight Profile Graph

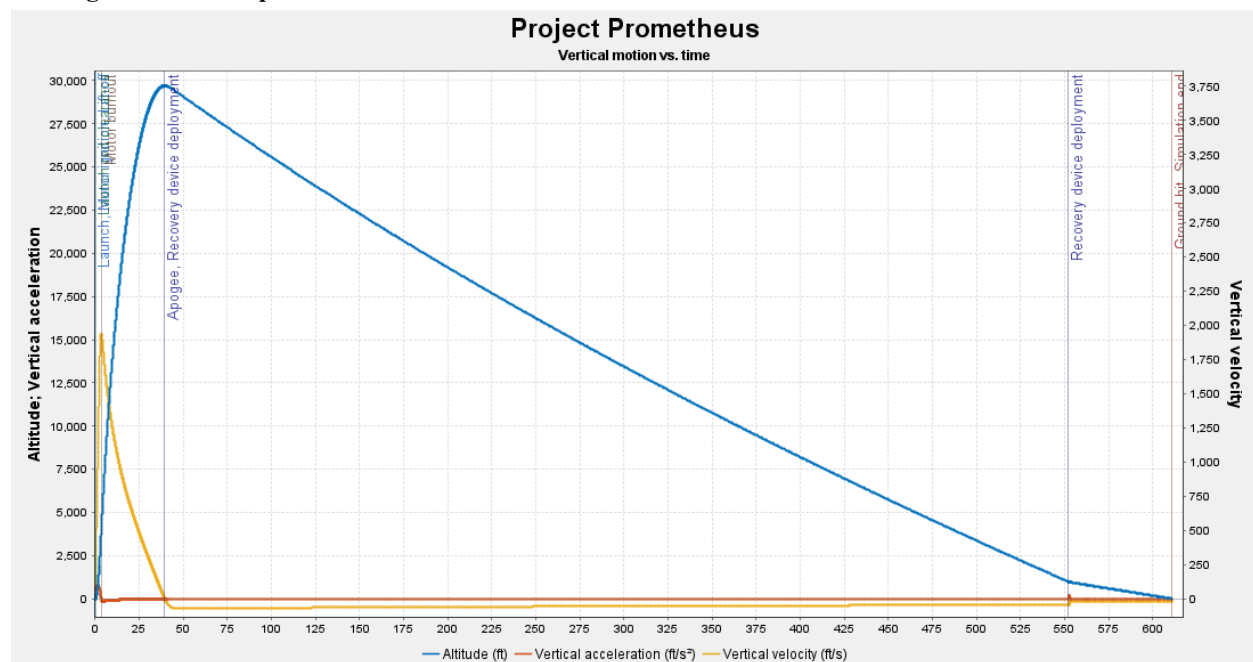


Fig. 3 Flight profile graph

E. Recovery Information

Table 4 Recovery Information

COTS altimeter	Telemega
Redundant altimeter	Easymini
Drogue primary deployment charge	16g CO2 canister
Drogue backup deployment charge	23g CO2 canister
Drogue descent rate	45 ft/s
Main primary deployment charge	7.5g black powder

Main backup deployment charge	9g black powder
Main deployment altitude	1000' AGL
Main descent rate	16 ft/s
Shock cords	25ft 2800lbs braided kevlar, 12ft 2100lbs tubular nylon webbing, 5/8 inch (x2 each)
Mechanical links	5/16" steel quick links, 3/8" steel eye bolts, 1000 lb swivel on drogue parachute, 3000 lb swivel on main parachute

Project Test Reports Appendix

A. Recovery System Testing

The recovery electronics were tested in a vacuum chamber to simulate the barometric pressure changes during flight. The e-matches, battery, and device were placed inside a vacuum bell jar and the pressure was reduced to about $\frac{1}{3}$ atmosphere, and then slowly increased to 1 atmosphere. As expected, drogue charges were fired as pressure began increasing, and main charges were fired once the barometric pressure matched the desired deployment altitude. The Telemega's estimated altitude and GPS position were also observed over radio link, and operated as expected.

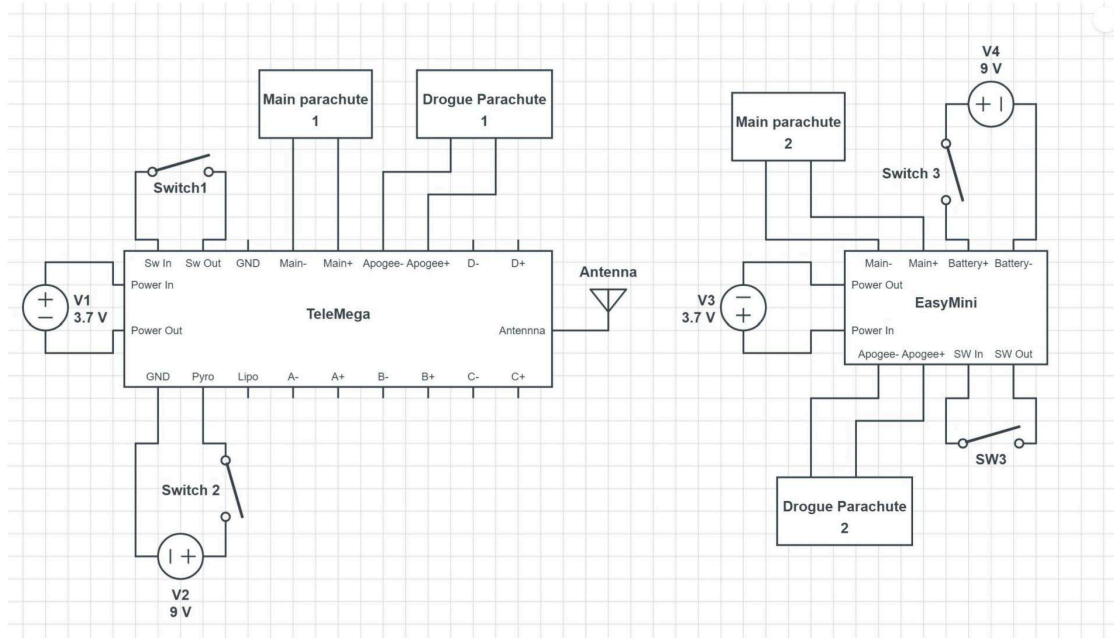


Fig. 4 Wiring diagram demonstrating dual redundancy of recovery electronics.

Validation that the separation charges were sufficient to induce the deployment of the drogue chute will be done in a separate test. The same amount of CO₂ as would be used for actual flight (16 grams) will be loaded into their charge canisters. Three nylon shear pins will be secured in pre-drilled holes in the rocket body. Ignition will be achieved using the same e-matches as would be used on a real flight, and will be triggered remotely by the actual flight computer. Flight will not occur unless this test is successful. If it is not, the quantity of CO₂ will be adjusted and re-tested.

A similar ejection charge test will be conducted for the main chute deployment. 7.5 grams of black powder will be loaded into a charge canister, then ignited remotely by the flight computer. Once again, flight will not occur unless this separation test is successful. If it is not, the quantity of black powder will be adjusted and re-tested.

B. SRAD Propulsion System Testing

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C. SRAD Pressure Vessel Testing

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D. SRAD GPS Testing

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E. Payload Recovery System Testing

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Hazard Analysis Appendix

Table 5 Hazard Analysis

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk of Injury after Mitigation
Igniter leads ignite during handling or storage, causing injury	Static electricity causing discharge, accidental connection to a battery or other voltage source	Moderate: igniters will be purchased on site and not handled until vehicle assembly	Ignited leads will be kept shorted until immediately before installation in the terminal blocks.	Low
Black powder or CO2 charges ignite during handling or storage, causing injury	Unsafe transport and storage, exposure to ignition source	Low: black powder and CO2 charges will be kept away from ignition sources while not in use	Powder and CO2 canisters will be kept in sealed container and ignition charge testing will be conducted to ensure no accidental ignitions	Low
Motor propellant ignites during handling or storage, causing injury	Unsafe transport and storage, Manufacturer error,	Low: Motor is a COTS and the probability of such an incident is low	Motor will be shipped by vendor to Spaceport America to minimize transportation risks. Ignitor will not be inserted until on the launch pad.	Low

Risk Assessment Appendix

Table 6 Risk Assessment

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk of Injury after Mitigation
Rocket deviates from nominal flight path, comes in contact with personnel at high speed	Strong winds present during launch, rocket unable to stabilize itself, manufacturing error	Low-Moderate: Rocket would not be launched during high winds, Severe deviation from nominal flight path is unlikely	Use longer launch rail to stabilize rocket quicker	Low
Recovery system fails to deploy, rocket or payload comes in contact with personnel	Poor electrical connections, battery failure, sensor failure	Moderate: The rocket features two commercial altimeters for redundancy which have been proven solutions	Ground testing of electrical connections, large enough battery to ensure battery life is not an issue	Low
Recovery system partially deploys, rocket or payload comes in contact with personnel	Sensor failure, poor electrical conditions	Low: parachute deployment is being controlled via the COTS altimeter and won't deploy unless armed	Keep recovery system unarmed until launch ready	Low
Recovery system deploys during assembly or prelaunch, causing injury	Sensor failure, poor electrical connections	Low: recovery system activates only under high acceleration and when height above 300 AGL is detected.	Avoid using excessive black powder. Calibrate the altimeter before arming. Check if the altimeter gives in built warning that 300 feet AGL has been reached when turned on and before armed	Low
Main parachute deploys at or near apogee, rocket or payload drifts to highway(s)	Sensor failure, poor electrical connections, accidental shear pin breakage	Low: recovery system has been tested to ensure apogee and drogue events do not deploy at the same time	3 4-40 nylon shear pins are used to prevent parachute from deploying due to shocks at apogee event	Low
Rocket does not ignite when command is given ("hang fire"), but does ignite when team approaches to troubleshoot	Motor malfunction, Improper setup	Moderate: Motor is a COTS so is expected to work, cause would be due to improper setup	Ensure motor is properly set up by following the launch checklist	Low

Rocket falls from launch rail during prelaunch preparations, causing injury	Strong winds, launch lugs fail or improperly placed	Low: launch would not be performed in strong winds. Rail buttons are reinforced with wood to increase failure loads.	Ensure that rail buttons have been properly secured with screws and epoxy	Low
Firewire e-matches ignite during assembly while loaded with black powder, causing injury	Poor electrical connections, accidentally activated by electronics	Moderate: When charge wells are loaded with black powder, electronics will be depowered and ignitors already connected to terminal blocks	Will not load black powder wells until just before final avionics integration	Low
Igniters ignite during assembly while not loaded with black powder, causing injury	Poor electrical connection, accidentally connected to electrical power source	Low: igniter leads will not be connected to electrical power source until rocket is launch ready	Keep ends covered and shorted until ready to connect to altimeters	Low

Assembly, Preflight, Launch, Recovery, and Off-nominal Checklists Appendix

A. Rocket Assembly

- A.1. Assemble payload (see Payload Assembly Checklist)
- A.2. Present payload to LCO payload inspection
- A.3. Mount payload into upper airframe.
- A.4. Assemble recovery electronics (see Recovery Electronics Assembly Checklist)
- A.5. Cut 4 e-matches to length and strip their leads, keeping the leads shorted to prevent ignition
- A.6. Before crimping e-match leads, test their resistance using the multimeter to ensure they are not shorted.
Nominal resistance is 1 ohm +/- 0.2 ohms
- A.7. Temporarily power electronics to check for ignitor continuity. Toggle one switch such that the secondary altimeter is heard to beep, wait approximately 1.5 seconds, and then toggle the other switch fully to arm the second altimeter. This is to ensure the continuity beeps do not overlap.
- A.8. If all continuity is reported on all 4 ignitors continue, if not, check wiring and restart.
- A.9. Disarm electronics
- A.10. Ensure quick link is attached to nosecone bulkplate.
- A.11. Assemble CO2 ejection mechanism, insert e-matches.
- A.12. Insert CO2 canisters for drogue parachute deployment: 16g for primary and 23g for secondary.
- A.13. Secure both CO2 ejection systems onto the upper airframe bulkhead with 1/4" bolts.
- A.14. Pack drogue parachute (see Recovery Assembly Checklist)
- A.15. Ensure that two sections of shock cord (tubular nylon and kevlar) are connected with a quick link.
- A.16. Attach drogue parachute to drogue shock cord.
- A.17. Attach shorter length of the drogue shock cord to eye bolt on nosecone bulkhead with quick-link.
- A.18. Insert drogue parachute into nosecone.
- A.19. Z-fold drogue shock cord and insert into nosecone.
- A.20. Integrate nosecone with upper body tube, secure with 3 4-40 nylon shear pins. Ensure centerlines of both parts are aligned.
- A.21. Pack main parachute (see Recovery Assembly Checklist)
- A.22. Attach main parachute to main shock cord
- A.23. Attach longer side of the shock cord to eyebolt on the motor forward closure.
- A.24. Insert main parachute into middle airframe.
- A.25. Z-fold main shock cord and insert into middle airframe
- A.26. Insert remaining main parachute length into middle airframe.
- A.27. Fill charge wells with black powder: 7.5 grams primary main, 9 grams secondary main.
- A.28. Insert e-matches into charge wells.
- A.29. Use dogbarf wadding to fill remaining space in charge wells, then use masking tape to seal wells.
- A.30. Use masking tape to cover bulkhead pass-through holes
- A.31. Insert avionics bay into upper/middle airframe coupler and secure with two 1/4-28 threaded rods, washers, and bolts.
- A.32. Attach lower airframe to middle airframe using 6 4-40 nylon shear pins.
- A.33. Attach middle airframe to avionics bay using 3 4-40 steel screws.
- A.34. Assemble AeroTech O5500X motor, with help from Animal Motor Works Rocketry and Team Mentor
- A.35. Insert motor into lower airframe motor mount and screw into motor retainer
- A.36. Present vehicle to LCO for final inspection. Move on to preflight checklist.

B. Parachute Folding

- B.1. Raise parachute into the air, making sure all shroud lines are straight.
- B.2. Lay the parachute on the ground.
- B.3. For the elliptical parachute:

- a. Fold so that all of the panels are folded into each other and the shroud lines come together at a point in the center of the parachute.
- b. Fold the shroud lines in a zig-zag line in the center of the parachute
- c. Fold the parachute into thirds parallel to the shroud lines
- d. Fold the parachute in half perpendicular to the shroud lines
- e. Attach swivel on parachute to quick link
- f. Attach flameproof shield to quick link
- g. "Burrito fold" parachute into flameproof protector

B.4. For the standard parachute:

- a. Lay it flat, then fold it in half so the shroud lines come together.
- b. Fold both sides into the middle parallel to shroud lines
- c. Fold the parachute in half again perpendicular to shroud lines
- d. Zig-zag the shroud lines back and forth in the center of the parachute
- e. Fold the main parachute in half yet again perpendicular to shroud lines - the loop at the top of the parachute should be seen
- f. Attach swivel on the parachute to quick link
- g. Attach flameproof protector to quick link
- h. "Burrito fold" parachute into flameproof protector

C. Recovery Electronics

- C.1. Install GPS tracker battery and secure with zip ties (battery) and 4-40 screws (tracker)
- C.2. Turn on GPS tracker and ensure transmission to ground station
- C.3. Double-check altimeter battery voltage with multimeter
- C.4. Install altimeter batteries (x2) and secure with zip ties
- C.5. Perform "tug test" - tug on all screwed-in connections by hand gently but firmly

D. Payload Assembly

- D.1. The payload will be presented fully assembled. Insert batteries on and ensure they are fully charged. Ensure SD card inserted and no metal to metal contact within CanSat. Create the tape cell used in electrodeposition experiment on the same day of the launch to minimize electrolyte evaporation.
- D.2. Power on Spot GPS tracker. Ensure good signal received.
- D.3. Power on Raspberry Pi on board the Electrodeposition experiment. Everything should be programmed to run automatically upon start-up.
- D.4. Install payload in the upper airframe with rods through the whole assembly. The CubeSats should be above the CanSat.

E. Preflight Checklist

- E.1. Load rocket onto rail. Ensure that there is no torque placed on rail buttons.
- E.2. Take promotional pictures
- E.3. Raise launch rail to an angle of TBD (nominally 84 degrees).
- E.4. Take promotional pictures
- E.5. Clear area of non-essential personnel
- E.6. Arm recovery electronics by turning both screw switches. Toggle one switch such that the secondary altimeter is heard to beep, wait approximately 1.5 seconds, and then toggle the other switch fully to arm the second altimeter. This is to ensure the continuity beeps do not overlap.
- E.7. Ensure that the electronics are reporting continuity in all 4 ignitors - 3 short beeps should be heard per altimeter. The primary altimeter will be set at a lower frequency and the secondary at a higher frequency. There will be 20 seconds between toggling and listening for continuity.
- E.8. Insert motor igniters - leave shunted for range manager to remove

F. Launch Checklist

- F.1. Wait for LCO to verify “go for flight” and direct student Team Lead to enter the launch command.
- F.2. Enter “FIRE” command into launch control unit
- F.3. In event of misfire or failed ignition, see off-nominal checklist
- F.4. In event of recovery systems deploying prematurely, see off-nominal checklist
- F.5. If flight is nominal, retrieve Consolidated Flight Card and Post Flight Record from LCO
- F.6. If the motor has a CATO, refer to off-nominal checklist

G. Recovery Checklist

- G.1. If rocket has landed in the vicinity of the launch site:
 - a. Recover rocket
 - b. Report to the MCO with rocket and Consolidated Flight Card and Post Flight Record
- G.2. If the rocket has not landed in the vicinity of the launch site:
 - a. Wait for LCO’s instructions to report to the MCO at a designated MCC with Consolidated Flight Card and Post Flight Record and recovery team members. The team will follow the IREC Range SOPs regarding recovery.

H. Off-Nominal Checklist

- H.1. Motor Misfire
 - a. Wait for LCO permission to re-enter the range
 - b. Remove motor igniters, check for root cause of error
 - c. Rectify error
 - d. Double-check igniter continuity as in checklist 6.7
 - e. Recycle to launch checklist
- H.2. Motor CATO
 - a. Wait for LCO permission to re-enter the range
 - b. Disarm recovery electronics by switching off the mechanical switch. If this is not possible, unscrew all e-matches to remove continuity, then remove recovery electronics from the main payload bay and disarm.
 - c. Perform any operations necessary to safe the motor remains
 - d. Bring rocket back to assembly area
 - e. De-power commercial GPS tracker
 - f. De-power payload
 - g. If damage is minor and will not affect flight, restart the assembly checklist. If damage is major, repair the rocket if possible then restart assembly checklist. If damage is critical and cannot be fixed with the given constraints, disassemble the rocket for transport.
- H.3. Accidental Deployment of Recovery System
 - a. Wait for LCO permission to re-enter the range
 - b. Remove igniters from the motor if still inserted
 - c. Disarm recovery electronics by turning off the mechanical switch. If this is not possible, unscrew all e-matches to remove continuity, then remove recovery electronics from main payload bay and disarm
 - d. Bring rocket back to assembly area
 - e. De-power commercial GPS tracker
 - f. De-power payload
 - g. If damage is minor and will not affect flight, restart the assembly checklist. If damage is major, repair the rocket if possible then restart assembly checklist. If damage is critical and cannot be fixed with the given constraints, disassemble the rocket for transport.

Equipment Checklist Appendix

A. Rocket Airframe

1. Lower airframe
2. Middle airframe
3. Avionics bay coupler
4. Upper airframe
5. Nosecone
6. Nosecone shoulder
7. Nosecone bulkplate
8. Payload bay bulkplate
9. Electronics bay bulkplate (x2)
10. 4-40 steel screws (x12)
11. 4-40 nylon shear pins (x12)

B. Propulsion

1. Motor: AeroTech O5500X (Animal Motor Works will bring to Spaceport America)
2. Motor casing: AeroTech casing, provided with kit
 - a. Main tube
 - b. Forward closure
 - c. Forward retaining ring
 - d. Aft retaining ring
 - e. Nozzle holder

C. Recovery System

1. Parachutes and associated systems
 - a. Main chute
 - b. Drogue chute
 - c. Drogue shock cord: tubular nylon
 - d. Drogue shock cord: kevlar
 - e. Main parachute deployment bag
 - f. Drogue parachute bag
 - g. Nomex squares (x4)
2. Hardware
 - a. Quick links (x8)
 - b. 2 ¼"-28 steel rods, ¼"-28 nuts (x24)
 - c. ⅜" Eye bolts, ⅜"-16 nuts, washers (x4)
 - d. Ejection charge canisters, ¼"-20 screws, ¼"-20 nuts (x6)
3. Recovery electronics
 - a. Flight computers (x2)
 - b. GPS module included on primary flight computer
 - c. Lithium-ion batteries (x3)
 - d. Flight computer to laptop wires
4. Black powder (7.5g, 9g)
5. CO2 gas canisters (16g, 23g)
6. Firewire e-matches (x4)
7. Wadding
8. 22 gauge stranded wire

D. Payload

1. CanSat
2. 2 CubeSats
3. Spot Tracker

4. Extra batteries
5. Extra MaxIQ spacers
6. Laptop with USB connection

E. Tools

1. Wire strippers and cutters
2. Soldering iron and solder
3. Screwdrivers
 - a. Hex wrench set
 - b. Small Phillips screwdriver set
 - c. Large Phillips screwdriver
4. Scissors
5. Sandpaper: 100, 150, 400 grit
6. JB Weld epoxy
7. Zip ties (small and large)
8. Masking tape
9. Electrical tape
10. Multimeter
11. Mixing cups and sticks
12. Pliers
13. Needlenose pliers
14. Blue loctite threadlock
15. X-acto knife

Engineering Drawings Appendix

We simulated the fincan in AeroFinSim and received the following results:

- Divergence Mach number: 3.89
- Flutter Mach number: 5.29
- Lift Slope CN-a: 16.7602

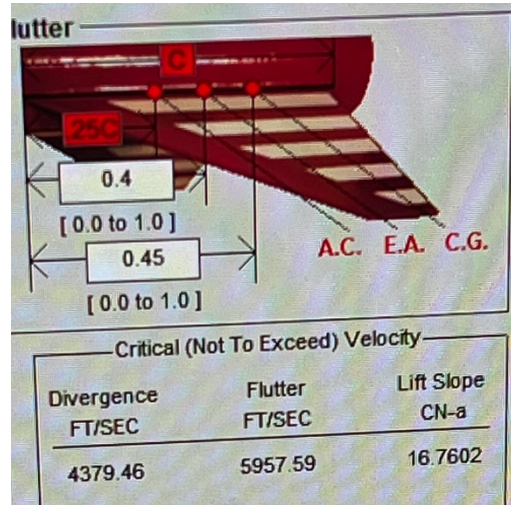


Fig. 5 AeroFinSim fin flutter analysis

Based on OpenRocket simulations, we do not expect the rocket to exceed Mach 1.8 anytime during flight, so fin flutter should not be an issue.

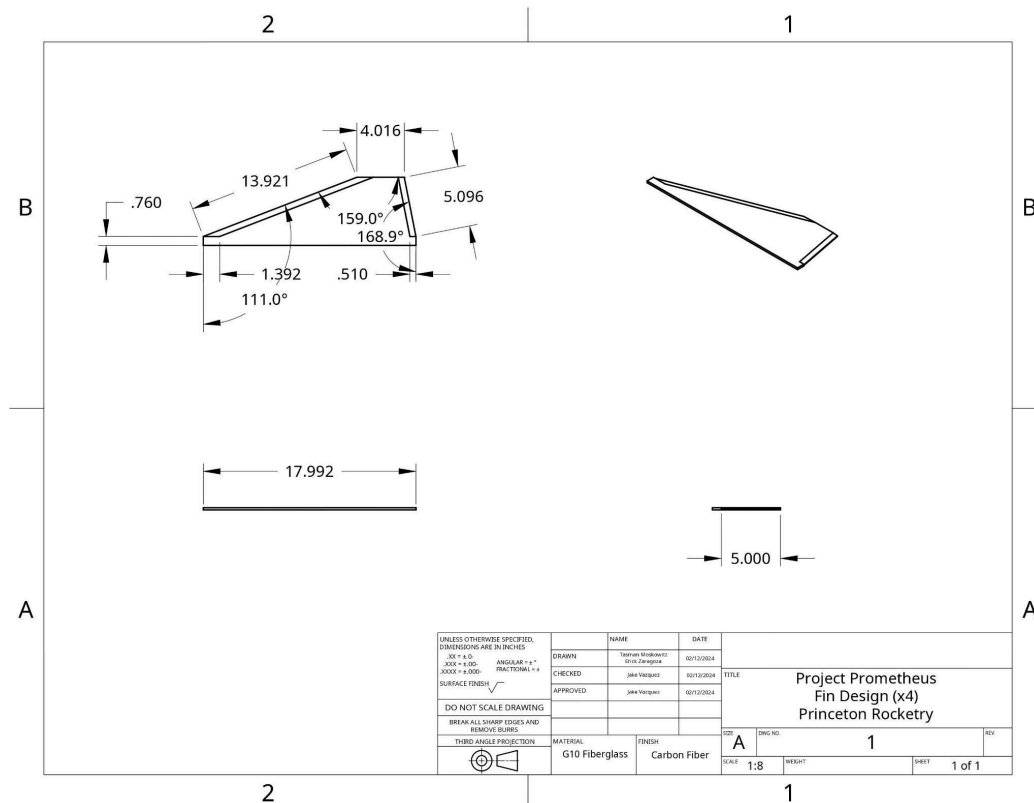


Fig. 6 Fin Design (x4)

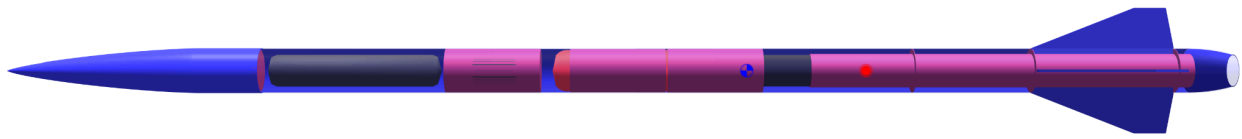


Fig. 7 Cutaway view of flight vehicle



Fig. 8 Render of flight vehicle

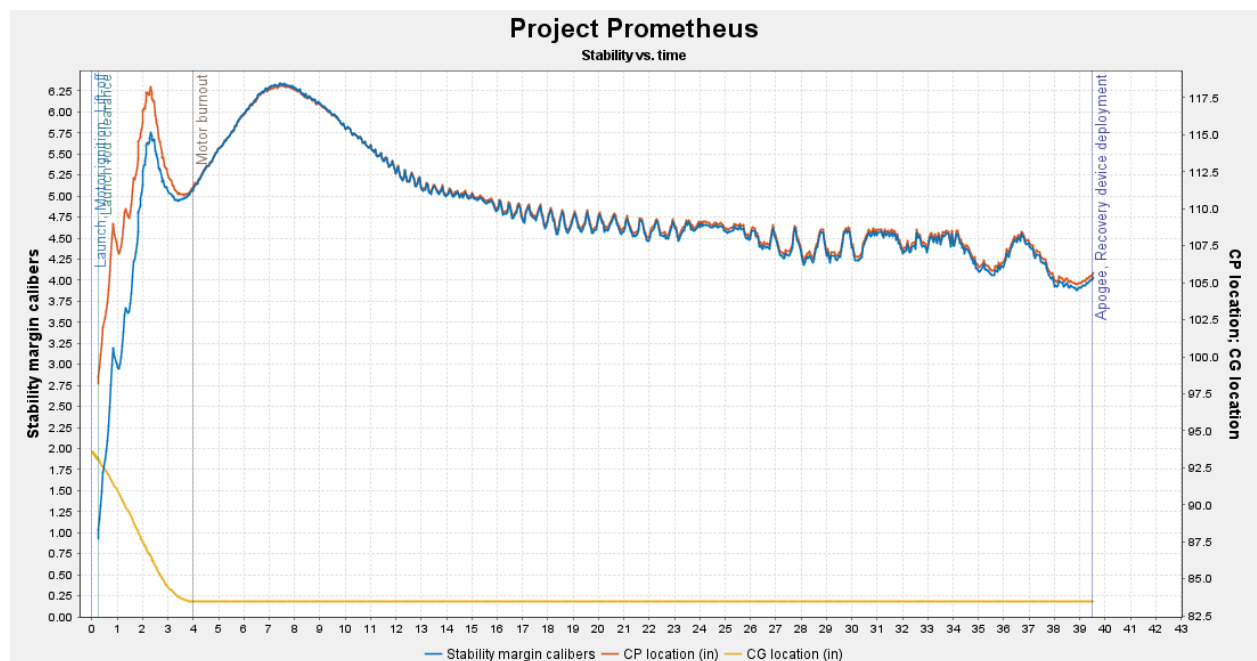


Fig. 9 Stability simulation of the flight vehicle

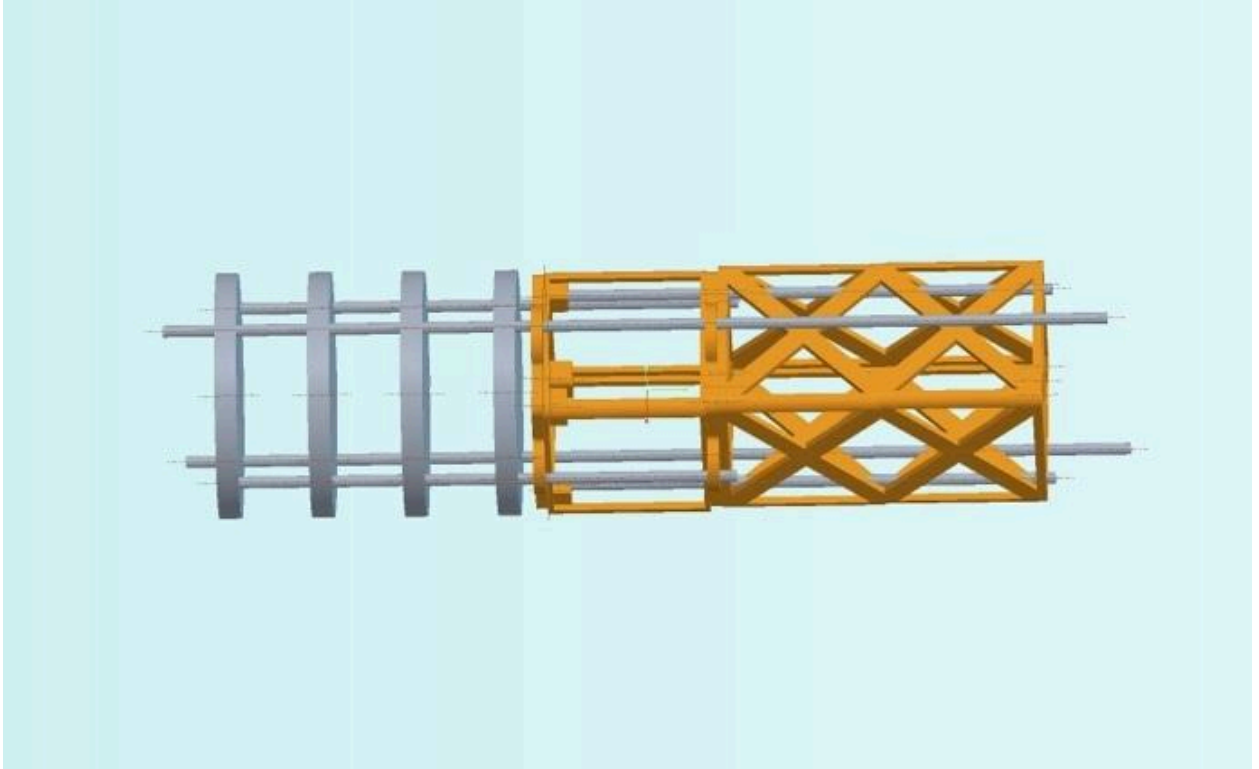


Fig. 10 Payload Assembly with CanSat and CubeSats

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